

1 Computing similarity transformations of operators

1.1 Motivation

Similarity transformations $B = P^{-1}AP$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form. Common examples include the Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. Here, we focus on similarity transformation of the form $B \mapsto e^A B e^{-A}$ with $A^\dagger = -A$ (generator anti-Hermitian). Hence, $U = e^A$ is unitary and the transformation preserves Hermiticity and spectra. Similarity transformations are also called equivalence or canonical transformation and play a big role in group theory.

1.2 Method A: The Hadamard lemma

Hadamard Lemma. *For any two operators A, B ,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \text{ad}_A^n(B), \quad \text{with} \quad (1)$$

$$\text{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \text{ad}_A^0(B) \equiv [A, B]. \quad (2)$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators. Utilize that for any A, X, Y ,

$$e^A (XY) e^{-A} = (e^A X e^{-A}) (e^A Y e^{-A}). \quad (3)$$

It thus suffices to compute $e^A a e^{-A}$ and $e^A a^\dagger e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut. If $A^\dagger = -A$, $U = e^A$ is unitary with $U^\dagger = e^{-A} = U^{-1}$, so $e^A B e^{-A} = U B U^\dagger$. Therefore, taking the dagger simply yields

$$(e^A B e^{-A})^\dagger = e^A B^\dagger e^{-A}. \quad (4)$$

Once $e^A a e^{-A}$ is known, $e^A a^\dagger e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

1.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A} X e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \text{ad}_A(X(\theta)). \quad (5)$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but it can be more tractable in practice.

Key fact: closure $\Rightarrow$ finite-dimensional linear ODE. If ad_A maps a finite set of operators $\{X_1, \dots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta} \mathbf{X}(\theta) = M \mathbf{X}(\theta), \quad [A, X_j] = \sum_i M_{ij} X_i, \quad \mathbf{X}(\theta) = e^{\theta M} \mathbf{X}(0), \quad (6)$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

1.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as $H_{\text{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\text{ad}_A(H_{\text{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\text{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A .

1.6 References

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